DATA DICTIONARY

The following data dictionary describes the fields, data types, keys, and constraints of the AgriTrack database based on the Entity Relationship Diagram (ERD).

1. USERS Table

Field	Data Type	Key / Constraint
id	INT	Primary Key (PK), Auto Increment
full_name	VARCHAR(100)	—
email	VARCHAR(150)	UNIQUE, NOT NULL
password_hash	VARCHAR(255)	—
role	ENUM('admin', 'manager', 'worker')	—
is_active	BOOLEAN	—
created_at	TIMESTAMP	—
updated_at	TIMESTAMP	—

2. FARMS Table

Field	Data Type	Key / Constraint
id	INT	Primary Key (PK), Auto Increment
name	VARCHAR(150)	—
location	VARCHAR(255)	—
area_hectares	DECIMAL(10,2)	—
owner_user_id	INT	Foreign Key (FK) → users.id
created_at	TIMESTAMP	—
updated_at	TIMESTAMP	—

3. FARM_MEMBERS Table

Field	Data Type	Key / Constraint
farm_id	INT	Composite PK, FK → farms.id
user_id	INT	Composite PK, FK → users.id
can_record_harvest	BOOLEAN	—
assigned_at	TIMESTAMP	—

The combination of farm_id and user_id forms the composite primary key. This table connects users to the farms they are assigned to.

4. CROPS Table

Field	Data Type	Key / Constraint
id	INT	Primary Key (PK), Auto Increment
name	VARCHAR(100)	NOT NULL
variety	VARCHAR(100)	Nullable
description	TEXT	Nullable
created_at	TIMESTAMP	—

5. PLANTINGS Table

Field	Data Type	Key / Constraint
id	INT	Primary Key (PK), Auto Increment
farm_id	INT	Foreign Key (FK) → farms.id
crop_id	INT	Foreign Key (FK) → crops.id
planted_area_hectares	DECIMAL(10,2)	—
planting_date	DATE	—
expected_harvest_date	DATE	Nullable
status	ENUM('planned', 'growing', ...)	Allowed values must be defined
created_at	TIMESTAMP	—
updated_at	TIMESTAMP	—

6. HARVESTS Table

Field	Data Type	Key / Constraint
id	INT	Primary Key (PK), Auto Increment
planting_id	INT	Foreign Key (FK) → plantings.id
harvest_date	DATE	—
quantity	DECIMAL(12,2)	CHECK (quantity >= 0)
unit	VARCHAR(20)	—
notes	TEXT	Nullable
recorded_by	INT	Foreign Key (FK) → users.id
created_at	TIMESTAMP	—

7. INVENTORY Table

Field	Data Type	Key / Constraint
id	INT	Primary Key (PK), Auto Increment
farm_id	INT	Foreign Key (FK) → farms.id
harvest_id	INT	Foreign Key (FK) → harvests.id; Nullable
item_name	VARCHAR(150)	—
quantity	DECIMAL(12,2)	—
unit	VARCHAR(20)	—
created_at	TIMESTAMP	—
updated_at	TIMESTAMP	—

Legend:

- **PK (Primary Key):** Uniquely identifies each record.
- **FK (Foreign Key):** Connects a record to another table.
- **UNIQUE:** Prevents duplicate values.
- **Nullable:** Allows a field to have no value.
- **Auto Increment:** Automatically generates an ID.
- **CHECK:** Enforces a specified condition on a value.
- **—:** No additional key or constraint is specified in the ERD.